Topology-Generated Potential (TGP):

Matter as the source of space — Executive summary

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April 2026

Abstract

Topology-Generated Potential (TGP) is a scalar-field theory in which **space is not a fixed background** but is generated by the presence of mass and energy. Each fundamental particle produces a local contribution to a shared field Φ, whose value measures the "density of generated space." Nonlinear interference of multiple sources produces three distinct regimes of interaction: (i) long-range attraction (gravity), (ii) intermediate-range repulsion, and (iii) short-range confinement. The fundamental constants c, $\hbar$, G are field-dependent, with the Planck length as the sole invariant.

The theory yields an exponential metric that reproduces all 10 PPN parameters of GR exactly ($\gamma = \beta = 1$), predicts $c_{\mathrm{GW}} = c_0$ identically (consistent with GW170817), and contains ΛCDM as a smooth limit ($\Delta\chi^2 = +1.8$, $\Delta\mathrm{AIC} = +3.8$ vs. ΛCDM on Planck + DESI data). Gravitational lensing is identical to GR in the weak field; deviations appear only in the strong-field regime (BH shadow $\sim$37% smaller). In the particle sector, a single substrate parameter $g_0^e = 0.869$ yields lepton mass ratios to 0.001% and $\alpha_s(M_Z) = 0.1174$ (0.6σ from PDG) with zero free parameters.

This document is a self-contained executive summary of the full 410-page manuscript [**?**]. Detailed derivations, proofs, and numerical verifications are available in the companion repository [**?**].

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1 Ontology

1.1 Central postulate

The Theory of Generated Space (TGP) rests on a single fundamental ontological claim:

Axiom 1 (Space from matter)**.** *Space does not exist independently of matter.* ***The presence of space follows from the presence of matter.*** *Every form of mass-energy generates a local contribution to a shared field Φ, which measures the "density of generated space" at a given location. Without matter there is no space; without space there is no matter. The two are mutually constitutive.*

This reverses the standard picture in which spacetime is a pre-existing arena and matter merely "moves" on it. It also distinguishes TGP from programmes in which space emerges from information or entanglement (AdS/CFT, ER = EPR): in TGP space emerges from *matter*.

Remark 2 (What TGP is *not*)**.** *TGP is* ***not*** *a scalar-tensor theory with an additional field living on a pre-existing manifold. The field Φ does not "inhabit" a prior spacetime — it* constitutes *that spacetime. A scalar-tensor formalism may arise only as an approximation in the late-time, weak-field limit of TGP, not as its foundation.*

Remark 3 (Three-level ontology of TGP)**.** *TGP adopts the following hierarchy:*

Level 0: Relational substrate $\Gamma = (V, E)$

Fundamental and non-emergent. Possesses adjacency relations and connectivity, but no metric. Exists in both sectors $\mathcal{S}_0$ and $\mathcal{S}_1$.

Level 1: Spatial field Φ (metric-phase parameter)

Φ is ***not*** *space or geometry — it is a* metric-phase parameter *of the substrate, measuring the degree to which the substrate supports metric relations (distances, propagation). $\Phi = 0$: the substrate exists but carries no metric (non-metric phase). $\Phi > 0$: metric phase — matter endows the substrate with the capacity to support geometry.*

Level 2: Effective geometry $g_{\mu\nu}$

Emergent: it arises from coarse-graining Level 1. Metric, curvature, and causality are properties of the configuration of Φ, not separate entities.

The substrate is fundamental and non-emergent. Geometry is a state of that substrate, controlled by Φ. The transition $\Phi = 0 \to \Phi > 0$ does not create space from nothing — it "switches on" the metric response of an already existing substrate. Analogy: freezing water does not create molecules; it changes their phase.

1.2 One substrate — many consequences

The three-level hierarchy above derives from a **single** fundamental object: the relational substrate $\Gamma = (V, E)$ with Hamiltonian H_Γ possessing a chiral $\mathbb{Z}_2$ symmetry (§1.6). Different coarse-graining projections of this one object yield:

1. **Field** $\Phi(x)$ — the metric-phase parameter, obtained from amplitude coarse-graining: $\Phi \propto \langle \hat{s}^2 \rangle_{\mathcal{B}(x)}$. It underpins the *entire* gravitational and cosmological sector (Sections 2–**??**).
2. **Field** $\sigma_{ab}(x)$ — the spatial-correlation tensor: $\sigma_{ab} \propto \langle \hat{s}_i \, \hat{s}_{i+\hat{e}_a} \rangle_{\mathcal{B}}^{\mathrm{TF}}$. It carries spin-2 and describes tensor gravitational-wave modes (§**??**). It follows from **the same** H_Γ as Φ and is not an additional postulate.

What this is *not*. The fields Φ and σ_{ab} do not live "on" space; they *constitute* space (Level 0 $\to$ 1 $\to$ 2). This is not a scalar-tensor theory: in a scalar-tensor theory two fields are *postulated* on a pre-existing manifold. Here both emerge from one substrate as different projections.

1.3 Nothingness N_0 and the state function $\mathcal{F}$

Axiom 4 (Nothingness). *The fundamental reference state is N_0 — absolute nothingness. Formally:*

$$N_0 := \{\emptyset,\ \mathcal{F}\}, \qquad \mathcal{F}\big|_{N_0} = 0, \qquad \Phi\big|_{N_0} = 0. \tag{1}$$

In the state N_0 there is no space ($\Phi = 0$), no matter, no time, no energy. The only logical property is that the state function $\mathcal{F}$ equals zero.

N_0 is not empty space — it is the absence of space. It is not the quantum vacuum — it is the absence of any structure.

Remark 5 (Substrate in N_0: terminological clarification). *In TGP the expression* absolute nothingness *means the* ***absence of all physical metric structure****: no space, matter, time, or propagation. It does* not *mean absence of the substrate Γ. The relational substrate is a Level-0 entity: it exists in both sectors $\mathcal{S}_0$ and $\mathcal{S}_1$. The state N_0 is the non-metric phase of the substrate — a state in which the $\mathbb{Z}_2$ symmetry is unbroken ($\langle \hat{s}_i \rangle = 0$ for every i), implying $\Phi = 0$ and absence of a metric.*

Statement	***In TGP***	***Justification***
"No metric in N_0"	*True*	$\Phi = 0$
"No matter in N_0"	*True*	$\langle \hat{s}_i \rangle = 0$
"No time in N_0"	*True*	*no propagation, $c \to \infty$*
"No substrate in N_0"	***False***	*Γ is fundamental*

Definition 6 (State function $\mathcal{F}$). *The state function $\mathcal{F}$ measures the* ***departure from nothingness****:*

- $\mathcal{F} = 0$*: full symmetry, no structure, state N_0.*
- $\mathcal{F} > 0$*: local symmetry breaking — matter exists together with the space it generates.*

At the level of the formalism, $\mathcal{F}$ is realised as the ***absolute state function*** $\mathcal{F}_{\text{abs}}(x) = \Phi(x)^2/\Phi_0^2$*, which vanishes exactly in N_0 ($\Phi = 0$).*

Axiom 7 (Zero-sum principle). *Viewed from outside, a closed system (the Universe) is indistinguishable from N_0:*

$$\mathcal{F}_{\text{ext}}\big(\textit{Universe}\big) = 0. \tag{2}$$

Internally $\mathcal{F}_{\text{int}}(x) > 0$ locally. The total net "amount of existence" is zero — the Universe is a self-cancelling fluctuation of nothingness.

1.4 Two sectors

Definition 8 (TGP sectors). *The configuration space Ω splits into two sectors:*

$\mathcal{S}_0 = \{\sigma : \mathcal{F}[\sigma] = 0\}$ — **nothingness sector**

No matter, no space, no time. The state N_0 belongs to this sector.

$\mathcal{S}_1 = \{\sigma : \mathcal{F}[\sigma] > 0\}$ — **existence sector**

Matter is present, space is generated, time exists as a relation between different states.

1.5 Instability of N_0

Axiom 9 (Instability of nothingness). *The state $\mathcal{F} = 0$ (N_0) is unstable. Its measure in configuration space vanishes:*

$$\mu\big(\{\sigma \in \Omega : \mathcal{F}[\sigma] = 0\}\big) = 0. \tag{3}$$

Any departure from exact symmetry leads to $\mathcal{F} > 0$ locally, entailing the simultaneous appearance of matter and the space it generates.

Proposition 10 (Instability from substrate thermodynamics). *Let the relational substrate $\Gamma = (V, E)$ have a Hamiltonian H_Γ with chiral $\mathbb{Z}_2$ symmetry, and let μ_β be the Gibbs measure $\mu_\beta(\sigma) \propto e^{-\beta H_\Gamma[\sigma]}$. In the thermodynamic limit ($|V| \to \infty$) there exists $\beta_c > 0$ such that*

$$\beta > \beta_c \quad \Longrightarrow \quad \mu_\beta\big(\{\sigma : \mathcal{F}[\sigma] = 0\}\big) = 0, \qquad \mu_\beta\big(\{\sigma : \mathcal{F}[\sigma] > 0\}\big) = 1. \tag{4}$$

The physical substrate (with no external thermostat) corresponds to $\beta \to \infty$ — it is always in the spontaneously broken regime. Axiom 9 is therefore a ***consequence*** *of substrate thermodynamics, not an independent postulate.*

1.6 Chiral symmetry of the substrate

On the relational substrate $\Gamma = (V, E)$, every node $i \in V$ carries an **amplitude operator** $\hat{s}_i$ — a scalar describing the local degree of excitation.

Proposition 11 ($\mathbb{Z}_2$ symmetry from microdynamics). *The substrate Hamiltonian*

$$H_\Gamma = \sum_i \left[\frac{\hat{\pi}_i^2}{2\mu} + \frac{m_0^2}{2}\hat{s}_i^2 + \frac{\lambda_0}{4}\hat{s}_i^4 \right] - J \sum_{\langle ij \rangle} A_{ij}\, \hat{s}_i \hat{s}_j \tag{5}$$

is invariant under ***chiral inversion****:*

$$\hat{s}_i \;\mapsto\; -\hat{s}_i \qquad \forall\, i \in V. \tag{6}$$

This discrete $\mathbb{Z}_2$ symmetry is the fundamental symmetry of the substrate. The state function $\mathcal{F}$ measures the degree of its breaking.

Corollary 12 (Relation $\mathcal{F} \leftrightarrow \langle \hat{s} \rangle$). *In the mean-field approximation:*

$$\mathcal{F}_i[\Omega] = \langle \Omega | \hat{s}_i | \Omega \rangle^2. \tag{7}$$

$\mathcal{F} = 0 \;\Leftrightarrow\; \langle \hat{s}_i \rangle = 0$ (unbroken $\mathbb{Z}_2$: no structure, nothingness). $\mathcal{F} > 0 \;\Leftrightarrow$ the substrate has "chosen a side" — spontaneous breaking of $\mathbb{Z}_2$ simultaneously generates matter and space.

2 The space-generating field

2.1 Definition of the field Φ

Definition 13 (Spatial field Φ). *Φ is a real scalar field whose value at a given point measures the* ***density of generated space*** *at that location. It is not a field on a ready-made manifold — it is the field that* constitutes *the manifold:*

- *$\Phi = 0$: no space (nothingness N_0),*
- *$\Phi > 0$: space exists, its "density" equals Φ,*
- *$\Phi \to \infty$: extremely dense space (interior of massive objects).*

Φ is not an **additional field** in space. It is the field **from which** space is composed. Metric, distance, causality — these are properties of the configuration of Φ, not separate entities.

Remark 14 (Physical interpretation). *One may think of Φ as a "density of aether" — but not of a material medium; rather, of the very* capacity for spatial relations to exist. *Where $\Phi = 0$ there is nothing — not even emptiness. Where $\Phi > 0$, objects can coexist, interact, and move relative to one another.*

2.2 Matter as a source of Φ

Every fundamental particle generates a contribution to the field Φ. This is not an "external" coupling (matter in space coupled to a field); it is a **constitutive act**: a particle *is* a local excitation that simultaneously produces the spatial field around itself.

Axiom 15 (Matter generates space)**.** *Each source of mass-energy ρ_a produces a contribution to Φ:*

$$\mathcal{D}[\Phi_a] = q_a \cdot \rho_a, \tag{8}$$

*where $q_a > 0$ is the **spatial generation constant** of particle type a (analogous to charge, but for space), and $\mathcal{D}$ is the operator governing the dynamics of Φ.*

Remark 16 (Generation constant q_a)**.** *The constant q_a specifies how strongly a given particle type generates space. It may depend on mass, spin, or other particle properties. It is a new physical parameter, intrinsic to TGP and absent from standard physics. In the homogeneous limit ($q_a = q$ for all types) the theory reduces to a one-parameter form.*

Remark 17 (Nature of the operator $\mathcal{D}$)**.** *The operator $\mathcal{D}$ in Eq. (8) is not a standard Laplacian on a pre-existing manifold — that would violate Axiom 1. It must be **internal to** Φ: defined by the very configuration of the field that constitutes space. Its explicit form is derived in Section* **??** *(Definition* **??***).*

For reference, the linear part of the operator reads

$$\mathcal{D}[\Phi]\big|_{\text{lin}} = \frac{1}{\Phi_0}\,\nabla^2\Phi, \tag{9}$$

while the full field equation includes the nonlinear self-interference term $\mathcal{N}[\Phi]$ (Definition **??***, Section* **??***):*

$$\mathcal{N}[\Phi] = \frac{\alpha}{\Phi_0}\,\frac{(\nabla\Phi)^2}{\Phi} + \beta\,\frac{\Phi^2}{\Phi_0^2} - \gamma\,\frac{\Phi^3}{\Phi_0^3}, \tag{10}$$

where $\alpha = 2$ (from variation of the action), $\beta, \gamma > 0$, $[\beta] = [\gamma] = \mathrm{L}^{-2}$.

2.3 Single source: spherical generated space

For an isolated, static point particle of mass M located at $\mathbf{x}_0$, the generated field has spherical symmetry:

$$\Phi_{\text{single}}(r) = \frac{q\,M}{4\pi}\,\phi(r), \qquad r = |\mathbf{x} - \mathbf{x}_0|, \tag{11}$$

where $\phi(r)$ is the **generation profile** — a function of distance from the source.

The profile $\phi(r)$ is **not** assumed a priori to be e^{-mr}/r (Yukawa) or $1/r$ (Coulomb). Its form follows from the full, nonlinear dynamics of Φ and must be **derived** from the equations of the theory. At the present stage one may only assume general properties:

1. $\phi(r) > 0$ everywhere (space is generated, never "negative"),
2. $\phi(r) \to \infty$ as $r \to 0$ (spatial density diverges at the source centre),
3. $\phi(r) \to 0$ as $r \to \infty$ (space vanishes far from the source),
4. $\phi(r)$ is non-monotonic — it possesses an inflection point or local extremum, giving rise to three regimes (Section 3).

2.4 Multiple sources: superposition and interference

In the presence of many particles the total field Φ is not a simple sum of individual contributions. It is the result of **nonlinear interference**.

2.4.1 Linear approximation (weak field)

In the weak-field limit ($\Phi \ll \Phi_{\text{char}}$, where Φ_{char} is the nonlinearity scale) the contributions superpose linearly:

$$\Phi(\mathbf{x}) \simeq \sum_i \Phi_i(\mathbf{x}) = \sum_i \frac{q_i\, M_i}{4\pi}\, \phi(|\mathbf{x} - \mathbf{x}_i|). \tag{12}$$

This approximation corresponds to the Newtonian limit and suffices at large scales.

2.4.2 Full nonlinear regime

In regions where contributions from many sources overlap strongly (dense matter clusters, stellar interiors, atomic nuclei), the field equation becomes **nonlinear**:

$$\mathcal{D}[\Phi] = \sum_i q_i \rho_i + \mathcal{N}[\Phi], \tag{13}$$

where $\mathcal{N}[\Phi]$ describes the **self-interference** of the field — the nonlinear coupling of generated space with itself. The explicit form of $\mathcal{N}[\Phi]$ is given in Eq. (10) above and in Definition **??** (Section **??**); the full field equation is stated in Theorem **??**.

The self-interference term $\mathcal{N}[\Phi]$ is responsible for:

- repulsion at intermediate scales (high spatial potential between nearby sources),
- a well at extremely small scales (confinement),
- amplification and cancellation of the field depending on the relative configuration of sources.

Remark 18 (Interference, not mere superposition)**.** *The word "interference" is deliberate. In wave physics, interference means that the result of superposition depends on* phase *— constructive and destructive effects can occur. In TGP the analogous role is played by the* ***field gradient*** *$\nabla\Phi$: two nearby sources generate between them a region of* higher *Φ than either produces alone (constructive overlap); but the net force on objects depends on the* gradient *of Φ, which can point either toward or away from the source, depending on the scale. This is the mechanism behind the three regimes (Section 3).*

2.5 From Φ to the metric

The metric is not a fundamental object. It is an **effective description** of the configuration of Φ. In regions where $\Phi > 0$ and varies smoothly, one can define an effective metric:

$$g^{\text{eff}}_{\mu\nu} = g_{\mu\nu}[\Phi, \partial\Phi, \dots]. \tag{14}$$

The exact form of this relation is an **open problem** in TGP. At the present stage one can state:

- the metric is defined only where $\Phi > 0$;
- a homogeneous $\Phi = \text{const} > 0$ yields a flat (Minkowski) metric;
- gradients of Φ generate curvature — this is gravity in TGP;
- large Φ with large gradients corresponds to a strong gravitational field.

Hypothesis 19 (Minimal metric)**.** *In the simplest ansatz the effective metric takes the form*

$$ds^2 = -\frac{c(\Phi)^2}{c_0^2}\, dt^2 + \frac{\Phi}{\Phi_0}\, \delta_{ij}\, dx^i dx^j, \tag{15}$$

where $c(\Phi)$ is the local speed of light (Section 4) and Φ_0 is the reference value. The spatial part is modulated by Φ/Φ_0 — more generated space means a larger "volume" available in a given region.

This metric form is a **working hypothesis**, not an axiom. Its consistency with observations must be verified. A corrected (exponential) form yielding correct post-Newtonian parameters is derived in Section **??**.

3 Three interaction regimes

The central prediction of TGP is the existence of **three distinct regimes** of interaction, arising from the interference of the Φ fields generated by different sources. These are not three separate forces — they are **one dynamics** of the field Φ that yields a different sign of the net force at different distance scales.

3.1 Mechanism: gradient of Φ as force

A test object moving in the field Φ experiences a force proportional to the gradient:

$$\mathbf{F} \propto -\nabla V_{\text{eff}}(\Phi), \tag{16}$$

where $V_{\text{eff}}(\Phi)$ is the effective potential felt by a particle in the spatial field.

Key observation: V_{eff} is **not a monotonic** function of the distance from the source. This follows from the nonlinear coupling $\mathcal{N}[\Phi]$ (Eq. 13). The effective potential between two sources has the form

$$V_{\text{eff}}(d) = \underbrace{-\frac{(q\Phi_0)^2 M_1 M_2}{4\pi\, d}}_{E_{\text{lin}}\text{: attraction (large } d)} + \underbrace{\frac{2\beta\,(q\Phi_0)^2 M_1 M_2}{(4\pi)^2\,\Phi_0\, d}\ln\frac{d}{r_0}}_{E_\beta\text{: repulsion (intermediate } d)} - \underbrace{E_\gamma(d)}_{\sim d^{-3}\text{: well (small } d)} \tag{17}$$

where $E_\gamma(d) > 0$ is a quartic term that dominates at small d. The crucial feature is the **structure**: three sign changes of the gradient, resulting from the different growth rates of the linear ($\sim 1/d$), cubic (repulsive), and quartic (well) terms as the sources approach each other.

Remark 20 (Explicit link to the field equation)**.** *The terms in the potential* (17) *follow directly from the field equation* (**??**)*:*

- $E_{\text{lin}} \sim -1/d$*: from the linear part* $\nabla^2\Phi/\Phi_0$ *after double volume integration between two Gaussian spheres;*
- $E_\beta \sim +1/d^2$ *(repulsion): from the cubic term* $\beta\Phi^2/\Phi_0^2$ *in* $\mathcal{N}[\Phi]$*; dominant when gradients and the value of* Φ *are moderately large;*
- $E_\gamma \sim -1/d^3$ *(well): from the quartic term* $-\gamma\Phi^3/\Phi_0^3$*; dominant at very large* Φ *(extremely close sources, Regime III).*

The exact decomposition (Eqs. **??**–**??***) is given in §***??***.*

3.2 Regime I: large scales — gravity

Proposition 21 (Gravity as an interference effect)**.** *At large scales* ($r \gg r_{\text{rep}}$*, where* r_{rep} *is the repulsion scale) the interference of the* Φ *fields generated by two material objects causes the gradient of* Φ *to point* ***toward the other object****. The objects are pushed together.*

Mechanism: each object produces a "hill" of Φ around itself. Two distant objects have between them a region where their fields overlap *constructively* — Φ between them is higher than on the outside. A test object "rolls" down the Φ gradient toward the other body.

This is **gravity** in TGP: not curvature of a pre-existing geometry, but a force arising from the constructive interference of generated space.

In the weak-field, large-distance limit one recovers Newton's law:

$$F_{\text{grav}} \simeq -\frac{G_{\text{eff}}\, M_1\, M_2}{r^2} \qquad (r \gg r_{\text{rep}}). \tag{18}$$

The constant G_{eff} is here a **derived quantity**, expressed in terms of the generation constant q and the parameters of the profile $\phi(r)$, not introduced by hand.

3.3 Regime II: intermediate scales — repulsion

Proposition 22 (Spatial repulsion). *At intermediate scales ($r_{\text{well}} < r < r_{\text{rep}}$) the spatial potential near particles is so high that the gradient of Φ points* ***away from the source****. Objects are repelled.*

Mechanism: in the immediate neighbourhood of a particle the field Φ attains very high values (the particle generates dense space around itself). When a second object approaches, it enters a region of *steep rise* in Φ — the gradient points outward, meaning a repulsive force.

This repulsion may account for:

- stability of matter (particles do not collapse onto each other),
- quantum effects (the uncertainty principle as a consequence of spatial repulsion at small scales),
- degeneracy pressure (a many-body effect in confined volume).

Remark 23 (Connection with the uncertainty principle). *In standard quantum mechanics the Heisenberg uncertainty principle ($\Delta x \cdot \Delta p \geq \hbar/2$) is an axiom. In TGP it may* follow *from spatial repulsion: attempting to localise a particle more tightly means entering a region of high Φ and strong gradient, which generates a "kick-back" — a physical source of uncertainty. The following proposition formalises this mechanism.*

Proposition 24 (Uncertainty principle from TGP spatial repulsion). *Let a particle of mass m be localised within a region of size Δx in the field $\Phi(r)$ generated by itself (self-coupling, Axiom 15). In Regime II (repulsion, $r_{\text{well}} < \Delta x < r_{\text{rep}}$) the field energy in the core volume $V_{\text{core}} \sim (4\pi/3)\Delta x^3$ is*

$$E_{\text{field}}(\Delta x) = \int_0^{\Delta x} 4\pi r^2 \, \frac{\beta\, \Phi_{\text{loc}}^2(r)}{\Phi_0^2} \, dr \;\approx\; \frac{\beta\,(qm)^2}{4\pi\, \Phi_0^2}\, \Delta x, \tag{19}$$

where $\Phi_{\text{loc}}(r) = qm/(4\pi r)$ is the field in the linear approximation. ***Key point****: the field energy grows linearly with Δx (it accumulates in an ever-larger core). The kinetic energy from momentum uncertainty $\Delta p \sim \hbar(\Phi)/\Delta x$ is*

$$E_{\text{kin}} \sim \frac{(\Delta p)^2}{2m} = \frac{\hbar(\Phi)^2}{2m\,(\Delta x)^2}. \tag{20}$$

Minimising $E_{\text{total}} = E_{\text{kin}} + E_{\text{field}}$ with respect to Δx:

$$\frac{dE_{\text{total}}}{d(\Delta x)} = 0 \quad \Longrightarrow \quad -\frac{\hbar(\Phi)^2}{m\,(\Delta x^*)^3} + \frac{\beta\,(qm)^2}{4\pi\, \Phi_0^2} = 0, \tag{21}$$

yielding the minimal localisation scale

$$\Delta x^* = \left(\frac{4\pi\, \Phi_0^2\, \hbar^2(\Phi)}{m\,\beta\,(qm)^2} \right)^{1/3}. \tag{22}$$

The corresponding momentum is $\Delta p^* = \hbar(\Phi)/\Delta x^*$, so the product equals

$$\Delta x^* \cdot \Delta p^* = \hbar(\Phi). \tag{23}$$

Hence the uncertainty bound:

$$\boxed{\Delta x \cdot \Delta p \;\geq\; \hbar(\Phi) = \hbar_0 \sqrt{\Phi_0/\Phi}.} \tag{24}$$

For $\Phi = \Phi_0$ (reference conditions): $\hbar(\Phi_0) = \hbar_0$, recovering the standard Heisenberg uncertainty principle ***as a dynamical result****, not an axiom.*

Status and scope of validity. *The derivation* (19)–(22) *is a* ***schematic order-of-magnitude proof*** *for the linear-field model ($\Phi_{\text{loc}} \sim qm/(4\pi r)$) in the weak-field regime. Full precision requires: (a) spinorial connection in TGP (Section* **??***), (b) quantisation of Φ with the substrate cutoff (Appendix* **??***).*

Corollary 25 (Variability of the uncertainty bound with Φ). *In a strong field ($\Phi \gg \Phi_0$, near a compact object): $\hbar(\Phi) = \hbar_0\sqrt{\Phi_0/\Phi} \ll \hbar_0$. The effective uncertainty bound is* ***suppressed****:*

$$\Delta x \cdot \Delta p \;\geq\; \hbar(\Phi)/2 \;\ll\; \hbar_0/2.$$

Matter in a strong field is closer to classical*: quantumness fades naturally without additional axioms.*

Proposition 26 (Substrate propagation speed $v_W(\chi)$). *Let $\chi \equiv \Phi/\Phi_0$ be the normalised spatial scalar. The covariant field equation of TGP,*

$$\Box_g \Phi \;=\; \mathcal{N}[\Phi], \tag{25}$$

linearised around a homogeneous background $\chi = \chi_0$, yields the substrate perturbation speed

$$v_W(\chi) \;=\; \frac{c_0}{\sqrt{\chi}}, \tag{26}$$

where c_0 is the speed of light in the reference state ($\chi = 1$). In particular: $v_W(1) = c_0$, $v_W(\chi) \to \infty$ as $\chi \to 0$ (vacuum region), $v_W \to 0$ as $\chi \to \infty$ (interior of $\mathcal{S}_\infty$).

Tensor GW speed*: $v_{\rm GW}^{\rm tensor} = c_0$ (the σ_{ab} sector, independent of χ) — consistent with the observation GW170817.*

3.4 Regime III: extremely small scales — confining well

Proposition 27 (Spatial well). *At extremely small scales ($r < r_{\rm well}$) the nonlinear self-coupling $\mathcal{N}[\Phi]$ creates a* ***potential well*** *— a region in which two sources are "glued" together in a common trough of the field Φ. The gradient of Φ once again points inward.*

Mechanism: when two sources are extremely close, their Φ fields overlap so strongly that the nonlinear terms $\mathcal{N}[\Phi]$ dominate and create a deep well — a region from which escape requires supplying significant energy.

This regime may account for:

- the strong nuclear interaction (quark confinement): quarks are "glued" together in a shared spatial well;
- formation of hadronic bound states;
- asymptotic freedom: inside the well the gradients are small (flat bottom), so particles behave as though free.

3.5 Matter as defects of the field Φ

In TGP a material particle is a **localised defect** of the field Φ — a radial kink.

Definition 28 (Radial kink). *A* radial kink *of the field Φ is a spherically symmetric, localised solution of*

$$\nabla^2\Phi - V'_{\rm eff}(\Phi) = 0, \tag{27}$$

satisfying the boundary conditions

$$\Phi(0) = \Phi_0 \neq \Phi_0, \qquad \Phi(r \to \infty) = \Phi_0. \tag{28}$$

Proposition 29 (Defect mass). *The mass associated with a radial kink of Φ is*

$$m_{\rm def} = 4\pi \int_0^\infty \left[\frac{1}{2}\left(\frac{d\Phi}{dr}\right)^2 + V_{\rm eff}(\Phi) - V_{\rm eff}(\Phi_0)\right] r^2\, dr. \tag{29}$$

The integral is finite owing to the boundary condition $\Phi(r \to \infty) = \Phi_0$ and regular behaviour of Φ at the origin.

3.6 Summary: effective-potential diagram

The effective potential $V_{\text{eff}}(r)$ between two sources exhibits the following structure:

Scale	∇V_{eff}	Effect
$r < r_{\text{well}}$	$\rightarrow$ inward	Well (confinement)
$r_{\text{well}} < r < r_{\text{rep}}$	$\leftarrow$ outward	Repulsion
$r > r_{\text{rep}}$	$\rightarrow$ inward	Gravity (attraction)

The shape of V_{eff} resembles a Lennard-Jones potential with an additional well at small scales — but its origin is fundamentally different: it arises not from empirical fitting, but from the nonlinear interference of the field Φ.

4 Dynamic fundamental constants

In standard physics c, $\hbar$, and G are universal constants. In TGP, where space is generated by matter, these "constants" become **fields dependent on the local density of generated space** Φ. The only truly constant quantity is the Planck length ℓ_{P}.

4.1 Rationale: constants describe propagation

c sets the speed of light propagation, $\hbar$ the scale of quantum effects, G the strength of gravitational interaction. All three characterise properties of the *space in which propagation occurs*. Since space in TGP is not a fixed background but a dynamical field Φ, it is natural that the propagation properties—and hence the values of c, $\hbar$, G—depend on the local state of that field.

4.2 Speed of light $c(\Phi)$

Axiom 30 (Speed of light as a field)**.** *The speed of light is a local field dependent on Φ:*

$$c(\Phi) = c_0 \left(\frac{\Phi_0}{\Phi}\right)^{1/2}, \tag{30}$$

where c_0 is the observed speed of light at the reference density $\Phi = \Phi_0$.

Interpretation:
- The denser the generated space (larger Φ), the **slower** light propagates.
- $\Phi \to \infty$: $c \to 0$—propagation ceases ("frozen" space; black-hole interior).
- $\Phi \to 0$: $c \to \infty$—unphysical, since $\Phi = 0$ means the absence of space.

Dense space slows light—analogous to a material medium with a large refractive index, except that here no medium exists: the *fabric of space itself* is denser.

Remark 31 (Interpretation of $c(\Phi)$: local physical speed)**.** *In standard GR the locally measured speed of light is* always c_0 *(equivalence principle); variability of c appears only when* ***comparing different regions****.*

In TGP, Axiom 30 defines $c(\Phi)$ as the ***local physical speed of light****—the speed following from the effective metric $g^{\text{eff}}_{\mu\nu}(\Phi)$ at a given point:*

$$c_{\text{lok}}(\Phi) = c_0 \sqrt{\frac{\Phi_0}{\Phi}}. \tag{31}$$

An operationally local *observer, using instruments made of matter immersed in the same Φ, measures c_0 (self-consistency), because all measurement standards (c, $\hbar$, G) scale coherently*

with Φ in a manner that preserves ℓ_{P}. The variability of $c(\Phi)$ is therefore relational*: it is revealed only when comparing regions of different Φ.*

This distinguishes TGP from standard GR, where the variable coordinate speed is an artefact of the coordinate choice. In TGP $c(\Phi)$ is a ***physical*** *consequence of differing densities of generated space. The difference is testable: in TGP the ratio $c(\Phi_1)/c(\Phi_2)$ is* ***observable*** *and depends on Φ_1/Φ_2; in GR the analogous ratio depends on* curvature*, not on any scalar field.*

4.3 Planck constant $\hbar(\Phi)$

Axiom 32 (Emergent Planck constant)**.** *The effective Planck constant is a field:*

$$\hbar(\Phi) = \hbar_0 \left(\frac{\Phi_0}{\Phi}\right)^{1/2}, \tag{32}$$

where $\hbar_0$ is the observed Planck constant.

Interpretation:

- Dense space ($\Phi \gg \Phi_0$): $\hbar \to 0$—quantum effects are suppressed. Matter behaves classically even at the level of individual particles.
- Dilute space ($\Phi \sim \Phi_0$): $\hbar \sim \hbar_0$—standard quantum mechanics.

This yields **two paths to the classical limit**:

1. *Statistical path*: $N \to \infty$ (many particles; relative fluctuations vanish).
2. *Spatial path*: $\Phi \to \infty$ (dense space suppresses quantumness).

Remark 33 (Black holes and quantumness)**.** *Near a black hole Φ is enormous, so $\hbar \to 0$: matter becomes* ***classical*** *in strong gravitational fields. This is the opposite of the standard expectation (in GR quantum effects should* grow *near the singularity). In TGP the classical singularity is* naturally removed*: matter cannot "condense to a point" because the surrounding space is too dense.*

4.4 Gravitational constant $G(\Phi)$

Axiom 34 (Dynamic gravitational constant)**.** *The gravitational constant is a field determined by the condition of Planck-length constancy:*

$$G(\Phi) = G_0 \cdot \frac{\Phi_0}{\Phi}, \tag{33}$$

where G_0 is the observed gravitational constant.

Interpretation:

- Dense space: G decreases—gravity is **weaker**; more space "dilutes" the interaction.
- Dilute space: G increases—gravity is stronger (but there is also less space).

4.5 Constancy of the Planck length

The Planck length is defined as

$$\ell_{\mathrm{P}}^2 = \frac{\hbar G}{c^3}. \tag{34}$$

Substituting the Φ-dependent expressions (30), (32), (33):

$$\ell_{\mathrm{P}}^2 = \frac{\hbar_0 \left(\frac{\Phi_0}{\Phi}\right)^{1/2} \cdot G_0 \left(\frac{\Phi_0}{\Phi}\right)}{\left[c_0 \left(\frac{\Phi_0}{\Phi}\right)^{1/2}\right]^3} = \frac{\hbar_0 G_0}{c_0^3} \cdot \left(\frac{\Phi_0}{\Phi}\right)^{1/2+1-3/2} = \frac{\hbar_0 G_0}{c_0^3}. \tag{35}$$

Theorem 35 (Constancy of the Planck scale)**.** *The Planck length is* ***invariant:***

$$\boxed{\ell_{\mathrm{P}} = \sqrt{\frac{\hbar_0\, G_0}{c_0^3}} = \mathrm{const},} \tag{36}$$

independently of the local value of Φ. It is the only truly fundamental scale of TGP.

This is an **internal consistency condition**: despite c, $\hbar$, G being dynamical, the scale of quantum gravity remains fixed. The Φ-dependences of $c(\Phi)$, $\hbar(\Phi)$, $G(\Phi)$ are not independent assumptions but **a single coherent condition** arising from the nature of the field Φ.

4.6 Uniqueness of the exponents

The axioms above postulate specific power laws: $c \propto \Phi^{-1/2}$, $\hbar \propto \Phi^{-1/2}$, $G \propto \Phi^{-1}$. We now show that these exponents are the **unique** ones consistent with three internal requirements of TGP.

Theorem 36 (Uniqueness of exponents)**.** *Let $c(\Phi) = c_0(\Phi_0/\Phi)^a$, $\hbar(\Phi) = \hbar_0(\Phi_0/\Phi)^b$, $G(\Phi) = G_0(\Phi_0/\Phi)^g$. Assume:*

(W1) ***Planck-length constancy:*** $\ell_{\mathrm{P}}^2 = \hbar G/c^3 = \mathrm{const}$.

(W2) ***Speed of light from the metric:*** *The antipodal bridge metric (Proposition* **??***) yields $a = 1/2$.*

(W3) ***Propagator consistency (universality of local physics):*** *The effective mass $m_{\mathrm{eff}} = mc/\hbar$ must be Φ-independent, giving $a = b$.*

Then $a = b = 1/2$ (from W2 and W3), and W1 gives

$$b + g - 3a = 0 \quad \Longrightarrow \quad g = 3a - b = 3 \cdot \tfrac{1}{2} - \tfrac{1}{2} = 1. \tag{37}$$

The ***unique*** *solution is $(a, b, g) = (1/2,\ 1/2,\ 1)$—precisely Axioms 30–34.*

More generally, consider any Planck combination $c^p\, \hbar^q\, G^r$ and ask when it is Φ-independent. The Φ-exponent is $-\frac{1}{2}\,p - \frac{1}{2}\,q - r$. Setting this to zero gives $p + q + 2r = 0$. The Planck length corresponds to $(p, q, r) = (-3, 1, 1)$, satisfying $-3 + 1 + 2 = 0$. This is the *unique combination* (up to overall powers) that is Φ-invariant.

4.7 Consequences

1. **No cosmological-constant problem** in the standard sense: G varies with Φ, so the "constant" expressed through G is not constant.
2. **Gravitational lensing**: in regions of higher Φ light travels more slowly—this produces lensing without invoking curvature as a separate concept.
3. **Gravitational redshift**: a photon travelling from higher to lower Φ accelerates (c increases), causing a redshift—recovering the GR result.
4. **Classical limit in strong fields**: near massive objects $\hbar \to 0$, removing the need for quantum gravity in the strong-field regime.

4.8 Observability of dynamic constants

Variations of $c(\Phi)$, $\hbar(\Phi)$, $G(\Phi)$ are **relational**: a local observer always measures the reference values c_0, $\hbar_0$, G_0, because her measurement standards (rulers, clocks, spectrometers) scale with Φ in the same way. Effects are detectable only by *comparing* two regions of different Φ.

Remark 37 (Principle of relational observability)**.** *In TGP* only ratios of quantities from different spatial regions are observable. *There is no objective "absolute" value of c, $\hbar$, or G—they are numerically equal to c_0, $\hbar_0$, G_0* by definition *for a local observer. TGP effects manifest as anomalies in standard scaling relations (Hubble diagram, galaxy sizes, neutron-star lifetimes).*

5 Dark energy

In standard cosmology dark energy is a cosmological constant Λ inserted by hand into the Einstein equations, or a quintessence field with an arbitrary potential. In TGP dark energy arises from the **dynamics of generated space itself**: condensation of matter amplifies fluctuations of the field Φ, which generate additional space.

5.1 Mechanism

Proposition 38 (Dark energy from Φ fluctuations)**.** *The field Φ undergoes fluctuations—both quantum (on small scales) and classical/turbulent (superposition of contributions from many sources on large scales). These fluctuations generate* ***additional effective space****, because the large-scale average $\langle\Phi\rangle$ is elevated relative to the value that the same mass would produce if distributed homogeneously.*

The detailed mechanism proceeds in four steps:

1. Every particle generates a field Φ in its vicinity.
2. When particles condense into structures (galaxies, clusters), their Φ fields superpose **nonlinearly**.
3. The nonlinearity $\mathcal{N}[\Phi]$ implies that the average $\langle\Phi\rangle$ in a structured region is **higher** than Φ for the same mass spread homogeneously:

$$\langle\Phi_{\text{struct}}\rangle > \Phi_{\text{homogeneous}}. \tag{38}$$

4. The excess $\Delta\Phi = \langle\Phi_{\text{struct}}\rangle - \Phi_{\text{homogeneous}}$ represents **additional space**—space that "should not exist" if matter were distributed uniformly. This additional space manifests as **accelerated expansion**: there is more "room" than matter alone would warrant, so the Universe expands faster than a matter-only model predicts.

Remark 39 (Flattening the problem)**.** *In standard cosmology one asks: "Why does Λ have* this *particular value?" (fine-tuning problem). In TGP the question becomes: "How much additional space do Φ fluctuations generate for a given matter distribution?" —the answer follows from dynamics, not from parameter tuning.*

5.2 Link to structure growth

A key prediction: dark energy in TGP **grows** as cosmic structures form.

- In the early Universe matter is nearly homogeneous, $\Delta\Phi \approx 0$, and dark energy is negligibly small.
- In the late Universe—galaxies, clusters, superclusters—matter is strongly concentrated, $\Delta\Phi$ is large, and dark energy is significant.

Hypothesis 40 (Coincidence problem)**.** *In TGP the "coincidental" overlap between the epoch of dark-energy domination and the epoch of structure formation* ***is not a coincidence****—one causes the other. Condensation of matter into structures generates additional space, which we observe as dark energy.*

5.3 Dark energy: natural Λ or quintessence?

Numerical analysis (script `w_de_exact.py`) reveals a central result. For **natural** parameters $\gamma \sim H_0^2/c_0^2$ (corresponding to $\tau_0 \sim 1/H_0$, $\Phi_0 \sim 25$), the field φ is **perfectly frozen**: Hubble friction $3H\dot{\varphi}$ dominates the potential force by ~ 40 orders of magnitude. The outcome is $w_0 = -1.000$, $w_a = 0.000$—**indistinguishable from a cosmological constant** Λ.

TGP is therefore a **natural Λ theory**: the cosmological constant is not a parameter inserted by hand but the residual potential $U(1) = \gamma/12$ (Proposition **??**).

The effective cosmological constant is computable:

$$\Lambda_{\text{eff}} \;=\; \frac{8\pi G_0}{c_0^4}\,\big\langle U(\varphi_{\min})\big\rangle_\Sigma, \tag{39}$$

where $\varphi_{\min}(\mathbf{x})$ solves the field equation subject to the zero-sum constraint, and $\langle\cdot\rangle_\Sigma$ denotes the spatial average.

When is dark energy dynamical? The field φ evolves appreciably ($w_0 > -1$, $w_a < 0$) only when $\gamma \gg H_0^2/c_0^2$, which requires either unnaturally large Φ_0, higher-order substrate couplings, or non-trivial initial conditions $\varphi_{\text{ini}} \neq 1$. In the dynamical regime TGP yields a **quintessence**-type prediction:

- $w_0 \gtrsim -1$: today $\dot\varphi^2 > 0$ but small.
- $w_a < 0$: as structures stabilise, $w_{\text{DE}} \to -1$ from above.

The equation of state of TGP dark energy takes the form (Proposition **??**):

$$w_{\text{DE}} \;=\; \frac{\frac{1}{2}\dot\varphi^2 - U(\varphi)}{\frac{1}{2}\dot\varphi^2 + U(\varphi)}. \tag{40}$$

In the slow-relaxation regime ($\dot\varphi^2 \ll U(\varphi)$):

$$w_{\text{DE}} \;\approx\; -1 + \delta w, \qquad \delta w \;=\; \frac{\dot\varphi^2}{U(\varphi)} \;>\; 0. \tag{41}$$

For natural $\gamma \sim H_0^2/c_0^2$ one obtains $\delta w < 10^{-40}$, operationally identical to $w = -1$. For enhanced γ: $\delta w \sim 0.02$–0.2, yielding testable quintessence with ($w_0 > -1$, $w_a < 0$) in the CPL parametrisation.

5.4 Relation to the zero-sum principle

The zero-sum principle (Axiom 7) requires $\mathcal{F}_{\text{ext}} = 0$. The additional space generated by fluctuations **does not violate** this principle:

- An internal observer sees more space (dark energy).
- An external observer sees $\mathcal{F} = 0$—the excess space is exactly compensated in the global balance.

Dark energy is therefore a **local self-organisation effect of matter**, not a global constant.

Remark 41 (Distinguishing TGP from ΛCDM)**.** *At natural parameters TGP is observationally indistinguishable from ΛCDM via $w(z)$ measurements alone. Discrimination requires **other observables**:*

1. ***GW polarisation**: a breathing mode alongside the tensorial $+/\times$ modes (Proposition **??**).*
2. ***Massive-GW dispersion relation**: a mass gap $f_{\text{cut}} \sim H_0/(2\pi)$.*
3. *Λ_{eff} **without fine-tuning**: $\Lambda \approx \gamma/12 \sim \Phi_0\, H_0^2/(12\, c_0^2)$ is* calculable*, not inserted by hand.*
4. ***Correlation of** Λ_{eff} **with structure**: the back-reaction mechanism (Section 5.2).*

6 Mathematical formalism

This section presents the core mathematical architecture of TGP in a self-contained form. The full derivations are given in the Polish-language manuscript (§8, §8a–c); here we state the principal results with their essential proofs.

6.1 The TGP kinetic operator

The dynamical variable of TGP is a single scalar field $\Phi(\mathbf{x}, t)$, interpreted as the coarse-grained density of the substrate lattice $\Gamma = (V, E)$. We work in the dimensionless variable $\varphi \equiv \Phi/\Phi_0$.

Theorem 42 (Uniqueness of $\mathcal{D}_{\rm kin}$)**.** *Among all local second-order scalar kinetic operators of the form $\mathcal{D}[\varphi] = K(\varphi)\,\nabla^2\varphi + \frac{1}{2}K'(\varphi)\,(\nabla\varphi)^2$, require:*

(C1) $K(\varphi)$ depends only on φ and is smooth for $\varphi > 0$ (locality);

(C2) $\mathcal{D}[\varphi] \to \nabla^2\varphi$ when $\varphi = \text{const}$ (Laplacian limit);

(C3) $\mathcal{D}$ is covariant under conformal rescaling $\varphi \mapsto \lambda\,\varphi$ of the substrate metric $h_{ij} = \varphi^4\,\delta_{ij}$ (Φ-covariance).

*Then $K(\varphi) = K_{\rm geo}\,\varphi^4$ (with $\alpha = 2$) is the **unique** solution, and the kinetic operator is*

$$\boxed{\mathcal{D}_{\rm kin}[\varphi] = \nabla^2\varphi + \frac{2\,(\nabla\varphi)^2}{\varphi} = \frac{1}{3\,\varphi^2}\,\nabla^2(\varphi^3).} \tag{42}$$

Proof sketch. (C2) fixes $K(1) = 1$. The Euler–Lagrange equation of $\mathcal{L}_{\rm kin} = -\frac{1}{2}K(\varphi)(\nabla\varphi)^2$ gives the ratio $\alpha(\varphi) = K'(\varphi)\,\varphi/[2K(\varphi)]$. (C1) requires $\alpha = \text{const}$. (C3) is checked by computing the Laplace–Beltrami operator on (h_{ij}): $\Delta_h\varphi = \varphi^{-4}\,\mathcal{D}_{\rm kin}[\varphi]$, which holds if and only if $\alpha = 2$. Hence $K(\varphi) = \varphi^4$. □

The identity $\mathcal{D}_{\rm kin} = (3\varphi^2)^{-1}\nabla^2(\varphi^3)$ shows that the operator is simply the flat Laplacian acting on φ^3, divided by $3\varphi^2$. This makes the nonlinearity transparent: it arises from the geometric coupling of the substrate lattice, not from an ad hoc modification of the field equation.

6.2 Lagrangian density and unified action

The complete TGP action is

$$S_{\rm TGP}[\Phi, \psi_m] = \int d^4x\,\sqrt{-g_{\rm eff}}\,\Big[\mathcal{L}_{\rm field}(\Phi) + \mathcal{L}_{\rm mat}(\Phi, \psi_m)\Big], \tag{43}$$

with three ingredients:

(a) Field Lagrangian.

$$\mathcal{L}_{\rm field} = -\frac{1}{2}\,K(\varphi)\,g_{\rm eff}^{\mu\nu}\,\partial_\mu\varphi\,\partial_\nu\varphi - U(\varphi), \tag{44}$$

where the kinetic function is

$$K(\varphi) = 1 + 2\alpha\,\ln\!\big(\varphi\big), \qquad \alpha = 2 \quad \text{(from 1-loop ERG)}, \tag{45}$$

and the self-interference potential is

$$U(\varphi) = \frac{\beta}{3}\,\varphi^3 - \frac{\gamma}{4}\,\varphi^4, \qquad \beta = \gamma \quad \text{(vacuum condition } U'(1) = 0\text{)}. \tag{46}$$

(b) Matter coupling.

$$\mathcal{L}_{\rm mat} = -\frac{q}{\Phi_0}\,\varphi\,\rho, \tag{47}$$

where ρ is the matter density minimally coupled to $g_{\mu\nu}^{\rm eff}$. The factor φ reflects the axiom that sources generate Φ, not a constant background.

(c) Volume element. From the effective metric (see §6.3 below):

$$\sqrt{-g_{\text{eff}}} = c_0\,\varphi. \tag{48}$$

Variation $\delta S/\delta\Phi = 0$ yields the full TGP field equation:

$$\boxed{\nabla^2\Phi + \frac{2\,(\nabla\Phi)^2}{\Phi} + \beta\,\frac{\Phi^2}{\Phi_0} - \gamma\,\frac{\Phi^3}{\Phi_0^2} = -q\,\Phi_0\,\rho.} \tag{49}$$

Remark 43 (TGP is not a scalar-tensor theory)**.** *The action* (43) *contains no Einstein–Hilbert term R. The effective metric $g^{\text{eff}}_{\mu\nu}$ is defined algebraically by Φ (it is not an independent dynamical variable). The Einstein equations emerge as a geometric consistency condition, not as a postulate. The only dynamical degree of freedom is Φ.*

6.3 Exponential metric from substrate coarse-graining

The spatial components of the effective metric follow directly from the substrate: a coarse-graining cell $B(\mathbf{x})$ containing N_B lattice nodes has physical volume $\propto \Phi(\mathbf{x})$. Hence:

$$g_{ij} = \frac{\Phi}{\Phi_0}\,\delta_{ij} = \psi\,\delta_{ij}, \qquad \psi \equiv \frac{\Phi}{\Phi_0}. \tag{50}$$

The temporal component is determined by the **information budget constraint**: the substrate has a fixed information capacity $\mathcal{B} = N_B\,s_0$ per cell, partitioned between spatial and temporal degrees of freedom. If spatial capacity scales as $h(\Phi) = \Phi/\Phi_0$ and temporal capacity as $f(\Phi)$, the multiplicative conservation law

$$f(\Phi)\cdot h(\Phi) = 1 \qquad \text{(antipodal condition)} \tag{51}$$

yields $f = \Phi_0/\Phi = 1/\psi$.

In the weak-field limit $\Phi = \Phi_0(1+2U/c_0^2)$, identifying the Newtonian potential U, the power-law form $g_{tt} = -c_0^2/\psi$ gives $\beta_{\text{PPN}} = 2$. The PPN consistency requirement $\beta_{\text{PPN}} = 1$ *uniquely* selects the exponential parametrization:

$$\boxed{ds^2 = -c_0^2\,e^{-2U/c_0^2}\,dt^2 + e^{+2U/c_0^2}\,\delta_{ij}\,dx^i\,dx^j, \qquad \psi = e^{2U/c_0^2}.} \tag{52}$$

Both forms ($1/\psi$ and e^{-2U/c_0^2}) satisfy $f\cdot h = 1$ and agree at $O(U)$; the exponential form extends the agreement to *all orders* in U.

Remark 44 (Derivation chain)**.** *The metric* (52) *is* ***not postulated*** *but derived from three inputs:*

(i) node density $\Rightarrow g_{ij} = \psi\,\delta_{ij}$;

(ii) information budget $\Rightarrow f\cdot h = 1$;

(iii) PPN consistency $\Rightarrow$ exponential form, $\gamma_{\text{PPN}} = \beta_{\text{PPN}} = 1$ exactly.

An independent check: requiring $\ell_P = \sqrt{G\hbar/c^3} = \text{const}$ (Axiom A7) fixes the scaling exponents $c(\Phi) = c_0/\sqrt{\psi}$, $G(\Phi) = G_0/\psi$, $\hbar(\Phi) = \hbar_0/\sqrt{\psi}$, all consistent with the exponential metric.

PPN parameters (exact). Expanding eq. (52):

$$g_{tt} = -c_0^2\big(1 - 2U/c_0^2 + 2U^2/c_0^4 - \ldots\big), \tag{53}$$

$$g_{ij} = \big(1 + 2U/c_0^2 + 2U^2/c_0^4 + \ldots\big)\delta_{ij}. \tag{54}$$

Comparing with the standard PPN expansion (Will 2014):

$$\gamma_{\text{PPN}} = 1, \qquad \beta_{\text{PPN}} = 1 \qquad \text{(exact to all orders in } U\text{)}. \tag{55}$$

6.4 Ghost resolution

The non-standard kinetic term $K(\varphi) = 1 + 4\ln\varphi$ vanishes at $\varphi^* = e^{-1/4} \approx 0.779$, signalling a potential kinetic ghost for $\varphi < \varphi^*$. Three observations resolve this:

1. Perturbative sector: ghost-free by construction. In the cosmological background $\psi_{\rm bg} > 0$, the Mukhanov–Sasaki kinetic coefficient is $Q_s = \psi_{\rm bg}^4 > 0$ for all physical $\psi_{\rm bg}$. In the fundamental variable ψ, the kinetic energy is $\psi^4(\nabla\psi)^2 \geq 0$.

2. Soliton sector: substrate resolution. The logarithmic function $K = 1 + 4\ln\varphi$ is a *truncated* approximation to the full substrate kinetic coupling:

$$K_{\rm sub}(\varphi) = K_{\rm geo}\,\varphi^4, \qquad K_{\rm geo} > 0, \tag{56}$$

which follows from the geometric link coupling $K_{ij} = J\,(\varphi_i\,\varphi_j)^2$ on the lattice. The expansion $K_{\rm sub} = K_{\rm geo}\,e^{4\ln\varphi} \approx K_{\rm geo}(1+4\ln\varphi+\ldots)$ shows that $K = 1+4\ln\varphi$ is its first-order logarithmic truncation. The full $K_{\rm sub} = K_{\rm geo}\,\varphi^4$ is **strictly positive** for all $\varphi > 0$:

$$K_{\rm sub}(\varphi^*) = K_{\rm geo}\,(e^{-1/4})^4 = K_{\rm geo}\,e^{-1} > 0. \tag{57}$$

The "ghost" at φ^* is an artefact of the continuum approximation, not a physical singularity.

3. Soliton equation: regular at all $\varphi > 0$. With the substrate coupling, the soliton ODE becomes

$$K_{\rm geo}\,\varphi^2\,\varphi'' + K_{\rm geo}\,\varphi\,(\varphi')^2 + \frac{2}{r}\,K_{\rm geo}\,\varphi^2\,\varphi' = V'(\varphi). \tag{58}$$

The coefficient of φ'' is $K_{\rm geo}\,\varphi^2 > 0$ for all $\varphi > 0$: no kinetic singularity, no sign change, no ghost.

Remark 45 (Ghost-freedom summary)**.** *TGP is ghost-free at three levels:*

Sector	***Kinetic positivity***	***Mechanism***
Fundamental (ψ)	$\psi^4(\nabla\psi)^2 > 0$	*geometric*
Perturbations (MS)	$Q_s = \psi_{\rm bg}^4 > 0$	*linearized*
Solitons (substrate)	$K_{\rm sub} = K_{\rm geo}\varphi^4 > 0$	*full lattice*

All three originate from the same substrate coupling $K_{ij} = J(\varphi_i\varphi_j)^2 \geq 0$, expressed in different variables and regimes.

6.5 Modified Friedmann equations

In the homogeneous FRW limit $\Phi = \Phi(t)$, defining $\psi(t) = \Phi(t)/\Phi_0$, the effective metric becomes

$$ds^2 = -\frac{c_0^2}{\sqrt{\psi}}\,dt^2 + a^2\sqrt{\psi}\,\delta_{ij}\,dx^i\,dx^j. \tag{59}$$

The TGP field equation in cosmological variables reads

$$\psi'' + \Big(3 + \frac{E'}{E}\Big)\psi' = \frac{\Phi_0}{E^2}\Big[\frac{7}{3}\,\psi^2 - 2\,\psi^3\Big] - \frac{2\,\psi'^2}{\psi}, \tag{60}$$

where primes denote $d/d\ln a$ and $E(a) = H(a)/H_0$.

The Einstein equations emerge as a *geometric consistency condition* of the effective metric. The modified Friedmann equation takes the form:

$$\boxed{3\left(H+\frac{\dot{\psi}}{4\psi}\right)^2\sqrt{\psi}=c_0^2\left[\frac{\sqrt{\psi}\,\dot{\psi}^2}{2\,c_0^2}+U(\psi)\right],} \tag{61}$$

with $U(\psi)=\frac{\gamma}{3}\psi^3-\frac{\gamma}{4}\psi^4$ (using $\beta=\gamma$).

In the vacuum ($\psi\approx 1$, $\dot{\psi}\approx 0$), this reduces to

$$3H_0^2\approx c_0^2\,U(1)=c_0^2\,\frac{\gamma}{12}=c_0^2\,\Lambda_{\rm eff}, \tag{62}$$

which is the standard de Sitter solution with effective cosmological constant $\Lambda_{\rm eff}=\gamma/12$. The dimensionless Friedmann equation $E^2(a)=\Omega_m/a^3+\Omega_r/a^4+\Omega_{\rm DE}\,f(\psi)$ with the TGP potential shape $f(\psi)=4\psi^3-3\psi^4$ recovers standard FRW cosmology, with effective dark energy arising from the scalar field potential.

Remark 46 (Bianchi identity propagation)**.** *The Friedmann constraint* (61) *is not obtained by varying the TGP action with respect to a (which gives an over-determined system), but is* propagated *by the contracted Bianchi identity* $\nabla_\mu G^{\mu\nu}_{\rm eff}=0$ *once it holds at the initial time* t_0*. This mirrors the role of the Hamiltonian constraint in GR.*

6.6 Parameter classification

TGP organizes its parameters into three layers:

Parameter	**Value**	**Layer**	**Origin**
Layer I — structural (universal across substrate cosmologies)			
$\alpha=2$	exact	I	$K_{ij}=J(\varphi_i\varphi_j)^2$
$\beta=\gamma$	exact	I	vacuum condition $U'(1)=0$
$d=3$	exact	I	lattice optimality
Layer II — selective (our universe)			
$\Phi_0\approx 24.66\ [H_0^2/c_0^2]^{-1}$	fitted	II	match to $\Lambda_{\rm obs}$
$a_\Gamma\approx 0.040$	fitted	II	match to $r_{21}=m_\mu/m_e$
Layer III — derived (predictions, not inputs)			
$c_{\rm GW}=c_0$	exact	III	substrate propagation
$\gamma_{\rm PPN}=\beta_{\rm PPN}=1$	exact	III	exponential metric
$\Lambda_{\rm eff}=\gamma/12$	$\sim H_0^2$	III	potential minimum
3 generations	exact	III	WKB $n=0,1,2$

The parsimony ratio is $M_{\rm obs}/N_{\rm param}\geq 12/2=6$: two free parameters ($\Phi_0$, a_Γ) generate at least twelve independent predictions. An open question (OP-3) is whether $a_\Gamma=1/\Phi_0$, which would reduce the free parameter count to one.

7 Observational tests

TGP has been confronted with a comprehensive suite of observational tests spanning the solar system, gravitational waves, gravitational lensing, large-scale structure, and background cosmology. This section summarizes the results; full derivations are given in the dedicated deliverables referenced below.

7.1 Parametrized Post-Newtonian analysis

(Full analysis: `nbody/tgp_ppn_full.tex`*, following Will (2014).)*

7.1.1 TGP as a metric theory

In TGP, matter couples universally to the effective metric $g_{\mu\nu}^{\text{eff}}$ and not directly to the scalar field Φ. Therefore TGP is a **metric theory** in the sense of Will (2014, §3.1). For any metric theory, six PPN parameters vanish identically:

$$\xi = \alpha_3 = \zeta_1 = \zeta_2 = \zeta_3 = \zeta_4 = 0. \tag{63}$$

7.1.2 γ and β from the exponential metric

From the exponential metric $ds^2 = -c_0^2\, e^{-2U/c_0^2}\, dt^2 + e^{+2U/c_0^2}\, \delta_{ij}\, dx^i dx^j$ (eq. 52), expanding to PPN order:

$$\boxed{\gamma_{\text{PPN}} = 1, \qquad \beta_{\text{PPN}} = 1} \qquad \text{(exact to all orders in } U\text{)}. \tag{64}$$

This is not an approximation: the exponential structure forces $\gamma = \beta = 1$ algebraically at every post-Newtonian order.

7.1.3 Preferred-frame parameters

For semi-conservative metric theories, the preferred-frame parameters α_1 and α_2 are proportional to $\partial\gamma/\partial\phi_0$ and $\partial(\gamma - \beta - 1)/\partial\phi_0$ (Damour & Esposito-Farèse 1992). Since $\gamma = 1$ and $\beta = 1$ are **exact** (independent of Φ):

$$\boxed{\alpha_1 = 0, \qquad \alpha_2 = 0.} \tag{65}$$

7.1.4 Complete PPN table

Parameter	GR	TGP	Derivation
γ	1	1	exponential metric (exact)
β	1	1	exponential metric (exact)
ξ	0	0	metric theory (automatic)
α_1	0	0	$\partial\gamma/\partial\phi_0 = 0$
α_2	0	0	$\partial(\gamma-\beta-1)/\partial\phi_0 = 0$
α_3	0	0	metric theory (automatic)
ζ_1	0	0	metric theory (automatic)
ζ_2	0	0	metric theory (automatic)
ζ_3	0	0	metric theory (automatic)
ζ_4	0	0	metric theory (automatic)

Result. TGP is **PPN-indistinguishable from GR**. All 10 parameters take their GR values *exactly*, not as a limit of some coupling constant.

Experimental bounds satisfied.

- Cassini: $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (Bertotti et al. 2003). TGP: $\gamma - 1 = 0$ exactly. Pass.
- Lunar Laser Ranging (Nordtvedt): $|4\beta - \gamma - 3| < 5.3 \times 10^{-4}$. TGP: $4\beta - \gamma - 3 = 0$ exactly. Pass.
- Binary pulsars: $|\alpha_1| < 1.3 \times 10^{-4}$, $|\alpha_2| < 1.8 \times 10^{-5}$. TGP: $\alpha_1 = \alpha_2 = 0$. Pass.

Caveat: strong-field regime. The PPN analysis is strictly valid for $U/c^2 \ll 1$. For neutron stars ($U/c^2 \sim 0.2$), strong-field "sensitivities" (Damour & Esposito-Farèse 1993) could modify the effective PPN parameters. Since TGP satisfies $\gamma = \beta = 1$, the Nordtvedt effect vanishes, but a fully rigorous strong-field analysis (solving the TGP field equation inside a neutron star) is left for future work.

7.2 Gravitational waves

(Full analysis: `nbody/tgp_gw_summary_table.tex`.)

7.2.1 Speed of gravitational waves

In TGP, gravitational and electromagnetic waves propagate through the *same* substrate Γ. The tensor perturbations σ_{ab} (spin-2) propagate on the background metric $g^{\rm eff}_{\mu\nu}$ independently of local fluctuations in Φ. Consequently:

$$\boxed{c_{\rm GW} = c_0} \qquad \text{(algebraic identity, not fine-tuning).} \tag{66}$$

This is consistent with the GW170817 + GRB 170817A constraint [?, ?] $|c_{\rm GW}/c - 1| < 3 \times 10^{-15}$.

7.2.2 Polarization content

TGP predicts:

- **2 tensor polarizations** (+, ×): massless, $c_{\rm GW} = c_0$, amplitude matched to GR via $\xi_{\rm eff} = 4\pi G_0 \sigma_0 \Phi_0$.
- **1 scalar breathing mode**: massive ($m_{\rm sp} = \sqrt{\gamma} \sim H_0$), with dispersion $\omega^2 = c_0^2(k^2 + m_{\rm sp}^2)$. Cutoff frequency $f_{\rm cut} \sim c_0\, m_{\rm sp}/(2\pi) \sim 10^{-18}$ Hz, far below the LIGO band. Non-detection in GW170817 is **expected** (screened).
- **No vector modes** in the minimal model.

The scalar breathing mode becomes detectable in the PTA band (∼nHz) with predicted amplitude $h_{\rm br} \sim 10^{-17}$, distinguishable from the stochastic GW background by its monopolar (isotropic) angular pattern. This is a **testable** prediction for SKA.

7.3 Gravitational lensing

(Full analysis: `nbody/tgp_lensing_geff.tex`.)

7.3.1 Weak-field deflection

For the TGP metric $\psi = e^{2U/c_0^2}$, the effective refractive index is $n(r) = \psi(r) = e^{2U/c_0^2}$, and $\ln\psi = 2U/c_0^2$ is *exact* (not a weak-field expansion). The Born integral closes analytically:

$$\hat{\alpha} = \int_{-\infty}^{+\infty} \frac{2\,G\,M\,b}{c_0^2\,(b^2 + l^2)^{3/2}}\, dl = \frac{4\,G\,M}{c_0^2\,b}, \tag{67}$$

identically the GR result. Numerical integration confirms this to $< 10^{-6}$ relative precision (8/8 tests PASS).

7.3.2 Shapiro time delay

The signal delay through a gravitational field is

$$\Delta t \approx \frac{4\,G\,M}{c_0^3}\, \ln\!\frac{4 r_1 r_2}{b^2}. \tag{68}$$

Linearization error: $O(U^2/c_0^4) \sim 10^{-12}$. The Cassini constraint $|\gamma - 1| = (2.1 \pm 2.3) \times 10^{-5}$ is satisfied automatically: TGP gives $\gamma - 1 = 0$ exactly.

7.3.3 Yukawa correction

The scalar perturbation $\delta\Phi$ around a point mass satisfies $(\nabla^2 - m_{\rm sp}^2)\,\delta\Phi = -q\,\Phi_0\,\rho$, yielding a Yukawa profile with effective coupling

$$\alpha_{\rm eff} = \frac{q\,\Phi_0}{4\pi} \approx 3.7 \times 10^{-26}. \tag{69}$$

The Compton wavelength $\lambda_{\rm sp} = 2\pi/m_{\rm sp} \approx 5.6\,$Gpc exceeds all astrophysical scales (unsuppressed Yukawa factor), but the amplitude is negligible:

$$\frac{\Delta\hat{\alpha}_{\rm Yuk}}{\hat{\alpha}_{\rm GR}} \sim \alpha_{\rm eff}^2 \sim 10^{-51} \qquad \text{(undetectable)}. \tag{70}$$

7.3.4 Post-post-Newtonian correction

The exponential metric deviates from Schwarzschild at $O(U^2)$: $g_{rr}^{\rm TGP} = 1 + 2U/c_0^2 + 2U^2/c_0^4 + \ldots$ versus $g_{rr}^{\rm Schw} = 1 + 2U/c_0^2 + 4U^2/c_0^4 + \ldots$. The correction enters only through non-Born ray-tracing effects:

$$\frac{\Delta\hat{\alpha}_{\rm ppN}}{\hat{\alpha}_{\rm GR}} \sim \left(\frac{GM_\odot}{R_\odot\,c_0^2}\right)^2 \approx 4.5 \times 10^{-12} \qquad (\sim 8 \text{ pico-arcsec}). \tag{71}$$

7.3.5 Strong-field prediction

Extrapolating the exponential metric to the strong-field regime:

Observable	**GR (Schwarzschild)**	**TGP (exponential)**	Δ/GR
Event horizon	$r_H = 2GM/c_0^2$	none (exp. > 0 always)	—
Photon sphere	$r_{\rm ph} = \frac{3}{2} r_s$	$r_{\rm ph} = r_s$	-33%
Shadow $b_{\rm crit}$	$3\sqrt{3}\,GM/c_0^2$	$r_s\sqrt{e}$	-37%

Caveat. These numbers assume the exponential metric extends to $U/c_0^2 \sim 0.5$. The actual TGP compact-object solution may differ; a complete strong-field lensing calculation is an open problem.

7.4 Cosmological perturbations and large-scale structure

(Full analysis: `nbody/tgp_perturbations_lss.tex`.)

7.4.1 Stability

Cosmological perturbations of Φ around the FRW background satisfy **all** stability conditions:

(i) **No-ghost**: $Q_s = \psi_{\rm bg}^4 > 0$ for all $\psi_{\rm bg} > 0$.
(ii) **Gradient stability**: $c_s^2 = c_0^2 > 0$ (sound speed equals light speed identically).
(iii) **Tachyonic stability**: $m_{\rm sp}^2 = \gamma > 0$ (exponential suppression on sub-Compton scales).

Verified numerically: 18/18 tests PASS over $\psi \in (0, 3]$.

7.4.2 Scale-dependent gravitational constant

The effective gravitational constant in the Fourier domain is

$$G_{\rm eff}(k, a) = G_0 \left(1 + \frac{2\,\alpha_{\rm eff}^2}{1 + (a\,m_{\rm sp}/k)^2}\right), \tag{72}$$

with $\alpha_{\rm eff} = q\,\Phi_0/(4\pi) \approx 3.7 \times 10^{-26}$. This gives:

$$\frac{G_{\rm eff}}{G_0} - 1 \sim 10^{-51} \qquad \Longrightarrow \qquad \text{TGP} = \Lambda\text{CDM in the LSS sector}. \tag{73}$$

7.4.3 Modified gravity parameterization (μ, Σ, η)

In the parametrized post-Friedmann framework:

$$\mu(k,a) = 1 + \frac{2\,\alpha_{\rm eff}^2}{1+(a\,m_{\rm sp}/k)^2}, \tag{74}$$

$$\Sigma(k,a) = 1 + \frac{\alpha_{\rm eff}^2}{1+(a\,m_{\rm sp}/k)^2}, \tag{75}$$

$$\eta(k,a) = \frac{\Sigma}{\mu} = \frac{1+\alpha_{\rm eff}^2/(1+x^2)}{1+2\alpha_{\rm eff}^2/(1+x^2)}, \qquad x = a\,m_{\rm sp}/k. \tag{76}$$

At $\Phi_0 \approx 25$: $\mu - 1 \sim \Sigma - 1 \sim \eta - 1 \sim 10^{-51}$, far below the projected sensitivity of Euclid ($\sigma_\mu \sim 0.02$) and LSST ($\sigma_\Sigma \sim 0.03$).

7.4.4 Inflationary predictions

The Mukhanov–Sasaki equation in TGP yields:

$$n_s = 1 - \frac{2}{N_e}, \qquad r = \frac{16}{N_e}, \qquad f_{\rm NL}^{\rm local} \approx -\frac{1}{N_e}, \tag{77}$$

where N_e is the number of e-folds. For $N_e \in [55, 65]$: $n_s \in [0.964, 0.969]$ (within Planck 1σ), $|f_{\rm NL}| \lesssim 0.02$ (undetectable).

Remark 47 (Note on r)**.** *The relation $r = 16/N_e$ assumes an unmodified tensor sector. In TGP, the tensor sector (σ_{ab}) has status "Sketch"; the actual $r_{\rm TGP}$ may differ once the full tensor perturbation theory is developed.*

7.5 Cosmological likelihood: TGP vs. ΛCDM

(Full analysis: `nbody/tgp_cosmo_likelihood.tex`.)

7.5.1 Data

Three complementary probes with $N_{\rm data} = 47$:

1. **DESI DR1 BAO** (DESI Collaboration 2024): D_V/r_d, D_M/r_d, D_H/r_d at $z_{\rm eff} = 0.30$–2.33 (11 points).
2. **Planck 2018 compressed CMB**: shift parameter R, acoustic scale ℓ_A, baryon density ω_b (3 points).
3. **Cosmic chronometers** $H(z)$ (Moresco et al. 2022): 33 differential-aging measurements.

7.5.2 Model specifications

ΛCDM (2 free parameters). Ω_m and H_0; $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$; $\omega_b = 0.02236$ fixed.

TGP (3 free parameters). Ω_m, H_0, and Φ_0. The dimensionless Friedmann equation is

$$E^2(a) = \frac{\Omega_m}{a^3} + \frac{\Omega_r}{a^4} + \Omega_{\rm DE}\, f\big(\psi(a)\big), \tag{78}$$

with the TGP potential shape $f(\psi) = 4\psi^3 - 3\psi^4$, $f(1) = 1$. Key limits:

- $\Phi_0 \to 0$: field frozen at $\psi = 1$, $\Rightarrow$ ΛCDM exactly.
- $\Phi_0 \sim O(1)$: quintessence-like $w_{\rm DE}(z)$.
- $\Phi_0 \gg 1$: rapid oscillations, $\langle w_{\rm DE}\rangle > -1$.

7.5.3 Results

Model	$\chi^2_{\rm BAO}$	$\chi^2_{H(z)}$	$\chi^2_{\rm CMB}$	$\chi^2_{\rm tot}$
ΛCDM ($\Omega_m = 0.305$, $H_0 = 68.0$)	14.3	24.0	2.1	40.3
TGP 3-param ($\Phi_0 \to 0$, free Ω_m, H_0)	15.1	24.2	2.9	42.1

Model	k	$\chi^2_{\rm tot}$	AIC	BIC
ΛCDM	2	40.3	44.3	48.0
TGP (3-param)	3	42.1	48.1	53.7
Δ (TGP $-$ ΛCDM)	+1	+1.8	+3.8	+5.7

Interpretation. $\Delta\chi^2 = +1.8$ with one extra parameter gives ΔAIC $= +3.8$: **weak preference for ΛCDM**, insufficient to distinguish the models. Both achieve $\chi^2/N < 1$ (adequate fits).

Key findings:

1. TGP **contains ΛCDM as a smooth limit** ($\Phi_0 = 0 \Leftrightarrow w_{\rm DE} = -1$); the 3-parameter best-fit converges to $\Phi_0 \to 0$.
2. Current data do not require $\Phi_0 \neq 0$: the field dynamics are not yet detectable.
3. TGP is **not excluded**: $\chi^2/N = 0.90$, comparable to ΛCDM ($\chi^2/N = 0.86$).
4. **Falsifiability**: TGP is falsified if future data require $w_{\rm DE}(z) < -1$ at any redshift (phantom crossing), since the TGP potential enforces $w_{\rm DE} \geq -1$ structurally.

7.6 Summary of observational status

Test	**TGP prediction**	**Status**
PPN (all 10 params)	= GR exactly	Pass
Cassini (γ)	$\gamma - 1 = 0$	Pass
LLR (Nordtvedt)	$4\beta - \gamma - 3 = 0$	Pass
$c_{\rm GW}$	$= c_0$ (algebraic)	Pass
GW polarizations	2 tensor + 1 screened scalar	Pass
Solar deflection	$\hat\alpha = 4GM/(c_0^2 b) =$ GR	Pass
Shapiro delay	$\gamma = 1$ exactly	Pass
Yukawa correction	$\Delta\hat\alpha/\hat\alpha \sim 10^{-51}$	Undetectable
ppN correction	~ 8 pico-arcsec	Undetectable
μ, Σ, η	$= 1 + O(10^{-51})$	$\equiv \Lambda$CDM
$f\sigma_8(z)$	$= \Lambda$CDM to $O(10^{-50})$	Pass
n_s	0.964–0.969 ($N_e = 55$–65)	Planck 1σ
BAO + CMB + $H(z)$	$\chi^2 = 42.1$ ($N = 47$)	Pass
BH shadow	$\sim 37\%$ smaller	Open
PTA breathing mode	$h_{\rm br} \sim 10^{-17}$	Testable (SKA)

In every currently measurable sector, TGP is operationally equivalent to GR (weak field) or ΛCDM (cosmology). The distinct predictions lie in: (a) the strong-field regime (BH shadow, frozen interior), (b) the PTA breathing mode, (c) the particle sector (lepton masses, α_s), and (d) N-body dynamics (Earnshaw violation, chaos suppression).

8 Predictions and falsifiability

8.1 Particle sector: the strongest argument for TGP

TGP is not merely a theory of gravity: as a theory of the scalar substrate from which the metric, fundamental constants, *and* particle masses emerge, TGP possesses a particle sector yielding quantitative predictions unmatched by any other modified-gravity theory.

8.1.1 Lepton masses from one parameter

The substrate soliton ODE

$$g^2\, g'' + g\,(g')^2 + \frac{2}{r}\, g^2\, g' = g^2(1-g) \tag{79}$$

has a one-parameter family of solutions labeled by the central value g_0. A single calibration parameter $g_0^e = 0.869$ (fixed by the muon-to-electron mass ratio $r_{21} = m_\mu/m_e$ via the φ-fixed-point mechanism) yields:

Observable	**TGP prediction**	**PDG value [?]**	**Accuracy**
$r_{21} = m_\mu/m_e$	$\left[A_{\text{tail}}(\varphi\, g_0^e)/A_{\text{tail}}(g_0^e)\right]^4 = 206.77$	206.768	0.001%
m_τ (via Koide $K = 2/3$)	1776.9 MeV	1776.86 MeV	0.002%

Seven additional lepton mass ratios are reproduced at $< 0.1\%$ precision from the same g_0^e.

8.1.2 Strong coupling constant: zero free parameters

The same g_0^e (determined purely from the lepton sector) predicts the QCD coupling at the Z-boson mass:

$$\alpha_s(M_Z) = \frac{N_c^3\, g_0^e}{8\, N_f^2} = \frac{27 \times 0.869}{8 \times 25} = 0.1174 \qquad (\text{PDG: } 0.1179 \pm 0.0009;\ 0.6\sigma). \tag{80}$$

This is a prediction with **zero** free parameters from the QCD or cosmological sectors; g_0^e comes entirely from lepton masses.

8.1.3 Discrete running: parameter-free test

The ratio of α_s at two flavor thresholds ($N_f = 3$ at m_τ; $N_f = 5$ at M_Z) is a **parameter-free** prediction:

$$\frac{\alpha_s(m_\tau)}{\alpha_s(M_Z)} = \left(\frac{5}{3}\right)^2 = \frac{25}{9} = 2.778 \qquad (\text{PDG: } 2.799 \pm 0.121;\ 0.18\sigma). \tag{81}$$

Remark 48 (Implication)**.** *The chain $r_{21} \to g_0^e \to \alpha_s(M_Z)$ constitutes a closed map: from one electroweak observable (lepton mass ratio), a QCD observable (strong coupling) follows. This is a feature unique to TGP, impossible in purely gravitational theories.*

8.2 Kill-shot tests (falsifiability)

TGP identifies four sharp falsification criteria, each targeting a structural prediction:

ID	**Observable**	**Falsification condition**	**Status**
K-A	c_{GW}	$c_{\text{GW}} \neq c_0$ at any level	**PASSED** (GW170817: $\lvert c_{\text{GW}}/c - 1\rvert < 3 \times 10^{-15}$)
K-B	γ_{PPN}	$\gamma_{\text{PPN}} \neq 1$ at any level	**PASSED** (Cassini: $\lvert\gamma - 1\rvert < 2.3 \times 10^{-5}$)
K-C	$G_{\text{eff}}(k)$	Scale-dependent G_{eff} at detectable level	**PASSED** ($\alpha_{\text{eff}} \sim 10^{-26}$; undetectable)
K-D	Lepton mass ratios	r_{21} or m_τ deviate from substrate ODE predictions	**PASSED** (0.001% and 0.002%)

Future kill-shots.

- **K-E**: Model-independent reconstruction confirming $w_{\rm DE}(z) < -1$ at $> 5\sigma$ at any redshift $\Rightarrow$ TGP falsified (the potential enforces $w_{\rm DE} \geq -1$). *Status*: DESI DR2 (2025) reports $w_0 > -1$, $w_a < 0$ at 2.5–3.9σ in CPL parameterisation. The apparent "phantom crossing" at $z \sim 0.4$ is an artifact of CPL: TGP naturally produces $w = -1$ (frozen ψ) at $z > 1$ and $w > -1$ ($\psi \to 7/6$ attractor) at $z < 1$—matching the DESI pattern **without** phantom crossing.
- **K-F**: Detection of $|\mu - 1| > 0.02$ or $|\Sigma - 1| > 0.03$ at fixed $\Phi_0 \approx 25$ (Euclid, LSST).
- **K-G**: $n_s \notin [0.95, 0.98]$ with no viable N_e (CMB-S4).
- **K-H**: Detection of sharp photon-ring substructure at $R_{\rm ph} = 3GM/c^2$ consistent with GR (ngEHT) $\Rightarrow$ TGP falsified. The soliton $\psi = (1 + 3r_s/r)^{1/3}$ has no photon sphere; the 1/3 exponent structurally forbids it for *any* lattice geometry (harmonicity protection, discrete Green's function regularization).

8.3 N-body dynamics: summary of 10 Results

(Full analysis: `dodatekY_nbody.tex` and `nbody/tgp_nbody_results_clean.tex`; 59/59 regression tests PASS, 104 active scripts.)

The N-body sector arises from the TGP scalar-field Lagrangian with quartic self-interaction $V(\varphi) = (\beta/3)\varphi^3 - (\gamma/4)\varphi^4$. The pairwise potential V_2 exhibits the three regimes of TGP (attraction, repulsion, confinement); the irreducible three-body force V_3 arises from the Φ^4 vertex. Ten principal results are:

1. **Earnshaw violation.** Equilateral 3-body configurations admit *stable static equilibria* with all normal modes $\omega^2 > 0$. In Newtonian gravity, Earnshaw's theorem forbids such equilibria; TGP evades it via the short-range repulsive barrier.
2. **Chaos suppression (35–45%).** The maximal Lyapunov exponent of the 3-body problem is reduced by 35–45% for $\beta \gtrsim 0.025$, enhanced by the three-body force V_3. The repulsive barrier prevents close encounters that drive chaotic divergence in Newtonian dynamics.
3. $C_{\rm eff}$ **from EFT.** The Yukawa source coupling $C_{\rm eff}$ is derived from an EFT projection of the classical topological defect (soliton) profile, connecting the microscopic substrate to the macroscopic N-body force law.
4. **Classical $\neq$ Yukawa.** The classical soliton–soliton interaction is *oscillatory* with power-law decay, qualitatively different from the Yukawa law $\propto e^{-m_{\rm sp} r}/r$ that requires the EFT mass gap.
5. I_Y **multipole representation.** The irreducible three-body Feynman integral I_Y admits a semi-analytic multipole expansion converging to $< 0.01\%$ at $L_{\max} = 15$.
6. V_3/V_2 **regime map.** The three-body force is perturbative ($\sim 1\%$) in the Lyapunov regime; exponentially suppressed for $m_{\rm sp}\, d > 3$.
7. **Analytical equilibria and Hill sphere.** Closed-form equilibrium distances $d_{\rm rep}$, $d_{\rm well}$; radial normal-mode frequency; TGP Hill sphere generically *expanded* relative to Newton.
8. **Unequal-mass phase diagram.** Phase boundary $\beta_{\rm crit} = \frac{3}{2}\sqrt{\gamma(C_1 + C_2)}$; escape velocity $v_{\rm esc} = \sqrt{2|V_2(d_{\rm well})|/\mu}$; breathing mode frequency $\omega_{\rm br}^2 = \frac{3}{2}\,\omega_{\rm rad}^2$ (exact), with V_3 correction of -1.3%.
9. N**-gon Earnshaw characterisation.** Full Hessian stability analysis of regular N-gons ($N = 2, \ldots, 8$) at the $d_{\rm well}$ equilibrium across a (β, C) grid. Result: Earnshaw violation persists for $N \leq 6$ at $\beta \leq 8$ and extends to $N \leq 8$ for $\beta = 20$. The critical polygon order $N_{\rm crit}(\beta)$ grows with coupling strength (wider confining well $\Rightarrow$ more bodies fit). For $N \geq N_{\rm crit}$, the farthest-pair distance $d_{\max} = 2R\sin(\pi\lfloor N/2\rfloor/N)$ exceeds the well, destabilising angular modes. The transition at $N_{\rm crit}$ may echo the three lepton generations from the soliton ODE (same confining-well geometry).
10. **Breathing mode time-domain verification.** Symplectic leapfrog integration of the

$N = 3$ breathing mode at $d_{\rm well}$ confirms the analytical frequency to 0.01% precision ($|E_{\rm rel}| < 2 \times 10^{-13}$, amplitude preserved to 10^{-6}). The N-body breathing mode is the localised analog of the cosmological PTA breathing mode ($A_{\rm br} \sim 8.6 \times 10^{-16}$): both originate from radial perturbations of Φ around Φ_0 and produce monopolar correlations.

Remark 49 (Cross-sector connections)**.** *The pairwise potential V_2 (gradient + repulsion + confinement) maps directly onto the three force regimes of TGP (§6). The three-body force V_3 arises from the Φ^4 self-interaction in the unified action (eq. 43). The EFT coupling $C_{\rm eff}$ provides the microscopic origin of the source strength C_i used throughout the N-body analysis, connecting to the soliton mass hierarchy ($r_{21} = 206.77$) of the particle sector. Three cross-sector links are now established: (i) the N-body breathing mode (Result 10) is the localised analog of the PTA scalar breathing mode (§8.4, item 3); (ii) the $N_{\rm crit}$ transition (Result 9) mirrors the three lepton generations from the same confining-well geometry of the soliton ODE (§8.1); (iii) $\alpha_s(M_Z) = 0.1174$ from g_0^e constrains $m_{\rm sp}$ and hence the screening length that sets the N-body force-law scale.*

8.4 Where TGP departs from GR

The PPN identity and the ΛCDM equivalence in the LSS sector do *not* mean TGP = GR. Deviations appear in five arenas:

1. **Cosmological dark energy:** Modified Friedmann equation with $G(\Phi)$ and $\Lambda_{\rm eff} = \gamma/12$; time-varying $w_{\rm DE}(z)$ controlled by Φ_0. Field frozen at $\psi = 1$ during matter era ($w = -1$), evolves toward attractor $\psi_{\rm eq} = 7/6$ in DE era ($w > -1$)—qualitatively consistent with DESI DR2 preference for $w_0 > -1$, $w_a < 0$ ($\Delta\chi^2 = +1.8$ vs. ΛCDM).
2. **Strong-field regime:** Frozen interior (no singularity) replacing the Schwarzschild/Kerr singularity. The substitution $u = \psi^3$ linearises the kinetic operator to $\nabla^2 u = 0$, yielding the exact BH soliton $\psi(r) = (1 + 3r_s/r)^{1/3}$. This solution has $g_{tt} \to 0$ at the centre but **no photon sphere**: $h(r) = r\,\psi(r)$ is strictly increasing (companion paper [**?**]).
3. **Scalar breathing mode:** Massive ($m_{\rm sp} \sim H_0$), screened in the LIGO band ($\lambda_{\rm sp} \sim 5500\,$Mpc), unscreened in the PTA band. Effective Brans–Dicke parameter $\omega_{\rm BD}^{\rm eff} = \Phi_0/4 \approx 6.25$ (Cassini bound evaded: $\gamma_{\rm PPN} = 1$ exactly). Predicted stochastic background: $A_{\rm br} \sim 8.6 \times 10^{-16}$ at $f = 1/{\rm yr}$ ($\sim 36\%$ of NANOGrav 15yr GWB amplitude), with **monopolar** angular correlations ($\Gamma(\theta) = $ const). Testable with existing PTA data via monopolar template reanalysis [**?**].
4. **Three-body forces and N-body stability:** V_3 from Φ^4 self-interaction, with no analogue in GR; Earnshaw violation for N-gons up to $N_{\rm crit}(\beta) \leq 8$, chaos suppression, expanded Hill sphere, breathing mode verified to 0.01%.
5. **Particle physics:** Lepton mass hierarchy from soliton ODE ($r_{21} = 206.77$, $m_\tau = 1776.9\,$MeV); $\alpha_s(M_Z) = 0.1174$ with zero free parameters.

These are the sectors where TGP can be *distinguished* from GR and ΛCDM by present or near-future experiments.

9 Relation to existing theories

TGP occupies a distinctive niche among extensions of GR. This section positions the theory relative to established frameworks and identifies points of contact and departure.

9.1 Scalar-tensor theories

The closest relatives of TGP are Brans–Dicke theory and its generalizations [**?**, **?**]. In these theories a scalar field φ is coupled to the Ricci scalar through $\omega(\varphi)R$, and the PPN parameters depend on ω. TGP differs in three structural ways:

1. The field Φ is not a conformal factor of an *a priori* metric; rather, the metric is an *emergent* object constructed from Φ.
2. The kinetic coupling $K(\varphi) \propto \varphi^4$ (with $\alpha = 2$) is not a free function but is determined by the uniqueness theorem for $\mathcal{D}$ and the vacuum condition $\beta = \gamma$.
3. The fundamental constants c, $\hbar$, G are all field-dependent, with $\ell_P = \text{const}$ as the sole invariant. This is more radical than VSL (variable speed of light) or VCC (varying coupling constant) models, yet avoids EEP violations because all dimensionless observables remain Φ-independent [?].

Despite these differences, TGP reproduces the GR limit with $\gamma_{\text{PPN}} = \beta_{\text{PPN}} = 1$ exactly [?], placing it in the same observational equivalence class as GR in the weak-field regime.

9.2 Gravitational wave constraints

The observation of GW170817 [?] with its electromagnetic counterpart constrains $|c_T/c - 1| \lesssim 10^{-15}$ [?], ruling out large classes of scalar-tensor and vector-tensor models. In TGP, $c_{\text{GW}} = c_0$ is an algebraic identity (the tensor perturbation propagates on the substrate, not on $g^{\text{eff}}_{\mu\nu}$), so this constraint is satisfied trivially.

9.3 Cosmological context

The Λ_{eff} mechanism in TGP is qualitatively similar to quintessence: a slowly evolving scalar potential provides an effective equation of state $w(z) \neq -1$. However, the physical origin is different: Λ_{eff} arises from *fluctuations of generated space amplified by matter condensation*, not from a fundamental potential. The likelihood analysis against Planck 2018 [?] and DESI DR1 [?] data gives $\Delta\chi^2 = +1.8$, $\Delta\text{AIC} = +3.8$ relative to ΛCDM — statistically indistinguishable.

9.4 Fuzzy dark matter analogy

The screened-Yukawa sector of TGP, with a Compton wavelength $\lambda_{\text{sp}} \sim$ Gpc, shares mathematical structure with fuzzy dark matter (FDM) [?, ?]. However, the effective boson mass in TGP ($m_{\text{sp}} \sim H_0$) is much larger than the typical FDM candidate ($m \sim 10^{-22}\,\text{eV}$). Current Lyman-$\alpha$ constraints [?, ?] do not exclude the TGP regime; future DESI Lyman-α data will provide a direct test (kill-shot K14).

9.5 Black hole thermodynamics

TGP reproduces the Bekenstein–Hawking entropy [?, ?] exactly: $S_{\text{BH}} = k_B A/(4\ell_P^2)$, with the Φ-dependence cancelling due to $\ell_P = \text{const}$. This is consistent with the Wald entropy [?] in the weak-field limit where the effective action reduces to Einstein–Hilbert. In the strong-field regime (BH shadow, singularity resolution), TGP makes predictions that differ from GR and are testable by the Event Horizon Telescope [?, ?].

9.6 Particle sector: Koide formula

The Koide relation $Q = (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2/(m_e + m_\mu + m_\tau) = 2/3$ [?] has been known as an empirical curiosity since 1982. In TGP, it arises as a *consequence* of the soliton fixed-point structure, not as an input. The single parameter g_0^e simultaneously determines lepton mass ratios (to 0.001%), the Koide angle, and $\alpha_s(M_Z)$ [?]. This is, to our knowledge, the first framework that derives $Q = 2/3$ from a dynamical mechanism rather than an *ad hoc* symmetry.

9.7 Reproducibility

All numerical results presented in this summary and in the companion papers are reproducible from the open-source repository [**?**], which contains 59 automated tests, a 10-stage master reproduction pipeline, and GitHub Actions CI running on Python 3.10/3.12.